AI-tool adoption and self-reported job satisfaction among professional software developers: a descriptive analysis of the Stack Overflow Annual Developer Survey

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Abstract

We use the Stack Overflow Annual Developer Survey 2025 to describe the cross-sectional relationship between professional developers' reported adoption of AI coding tools and their reported job satisfaction, controlling for years of professional work experience and primary developer role. Among the 24,203 respondents in our analysis sample, daily AI-tool users report a mean job satisfaction of 7.30 (out of 10), compared with 7.13 among respondents who do not use AI tools and do not plan to—a raw difference of 0.17 points. In a fully-specified OLS model that includes work-experience controls and developer-role fixed effects, AI-tool use (any vs. none) is associated with a satisfaction premium of 0.155 points ($SE = 0.033$, $p < 0.001$). The association is statistically significant but substantively small ($R^2 = 0.020$). The analysis is strictly descriptive: the survey is observational, selection into the survey is not random, and both AI-tool adoption and job satisfaction are self-reported. We discuss the limits these features impose on any causal interpretation. This paper is a teaching artefact produced for a seminar on AI for research; it is not intended for publication.

1 Introduction

Generative AI coding assistants—tools that autocomplete, explain, and generate code in response to natural-language prompts—have diffused rapidly through the software-development workforce since 2023. Whether adoption of these tools is associated with developers' wellbeing, as measured by self-reported job satisfaction, is an open empirical question. A positive association could reflect productivity gains that reduce frustration or increase perceived achievement; a negative association could reflect the anxiety of working alongside technology perceived as threatening or unreliable. We contribute a transparent, data-grounded look at this association using the largest annual survey of software developers.

The analysis is explicitly and entirely descriptive. We do not claim that AI-tool adoption causes any change in job satisfaction. The observational design, self-reported outcomes, and absence of plausible instruments for AI-tool adoption all rule out causal interpretation (see Section 5 for a detailed account). Our goal is to document the correlation, characterise its magnitude, and show that it survives controls for work experience and developer role.

This paper is a teaching artefact produced for a university seminar on AI for research. Its purpose is to demonstrate, end-to-end, how a researcher can move from a public dataset to a compiled, version-controlled manuscript using AI-assisted tooling—while maintaining the epistemic standards that separate a descriptive finding from a causal claim. The seminar workflow, the prompt templates, and all analysis code are available in the accompanying repository.

2 Data

We use the Stack Overflow Annual Developer Survey 2025 [Stack Overflow, 2025], an annual online survey distributed to registered Stack Overflow users and promoted through the platform's channels. The survey is released under the Open Database License (ODbL 1.0) and the Database Contents License (DbCL 1.0), permitting redistribution with attribution. Respondents are self-selected: participation requires awareness of the survey and willingness to complete it. The raw 2025 dataset contains 49,191 complete responses across 172 variables.

Analysis sample. We restrict the sample to respondents who self-identify as "a developer by profession" (37,467 individuals), then drop observations with missing values on any of the four variables required for the main regression—job satisfaction, AI-tool use frequency, years of professional work experience, and primary developer role—yielding a working sample of 24,354. We additionally exclude developer-role cells with fewer than 50 respondents to ensure stable fixed-effect estimates, giving a final analysis sample of **24,203 respondents**.

Key variables. *Job satisfaction* is measured on an integer scale from 1 (very dissatisfied) to 10 (very satisfied) (`JobSat`). The distribution in our sample has a mean of 7.22 and a median of 8, with a standard deviation of 2.0.

AI-tool use is measured by the survey item `AISelect`, which records the frequency with which a respondent uses AI tools: "No, and I don't plan to"; "No, but I plan to soon"; "Yes, monthly or infrequently"; "Yes, weekly"; and "Yes, daily". For the main binary specification we code any "Yes" response as uses_ai = 1 and both "No" responses as uses_ai = 0; 19,574 respondents (80.9%) report some AI use. For a robustness check we assign an ordinal score from 0 to 4 following the ordering above.

Work experience is measured in years at the respondent's current or most recent employer (`WorkExp`). The median in the analysis sample is 11 years and the mean is 13.6 years.

Developer role is the respondent's primary self-identified role (`DevType`, first listed category). The two most common roles are full-stack developer (25.1% of the analysis sample) and back-end developer (13.1%).

3 Method

We estimate the following OLS specification:

$$JobSat_i = \alpha + \beta_1\, UsesAI_i + \beta_2\, WorkExp_i + \beta_3\, WorkExp_i^2 + \boldsymbol{\gamma}'\mathbf{d}_i + \varepsilon_i \tag{1}$$

where $JobSat_i \in \{0, \ldots, 10\}$ is respondent i's reported job satisfaction, $UsesAI_i$ is the binary AI-adoption indicator, $WorkExp_i$ enters linearly and quadratically to allow a non-monotone experience profile, $\mathbf{d}_i$ is a vector of developer-role indicator variables (fixed effects), and ε_i is an idiosyncratic error term.

We estimate four nested models: (M1) the bivariate regression with $UsesAI_i$ only; (M2) adding the experience controls; (M3) the fully specified model with role fixed effects; and (M4) replacing the binary AI indicator with the ordinal 0–4 score to assess whether the relationship is monotone in AI-use intensity. All standard errors are heteroskedasticity-robust. Because the outcome is an integer bounded at 0 and 10, an ordered-probit model would be more appropriate in principle; we use OLS for expositional simplicity and note that results are qualitatively identical under an ordered specification.

β_1 in Model (M3) is our main descriptive parameter. It captures the difference in mean job satisfaction between AI-tool users and non-users, within developer role and conditional on the experience profile. It is a within-role, experience-adjusted correlation, not a causal effect.

4 Results

4.1 Descriptive patterns

Figure 1 plots mean job satisfaction and 95% confidence intervals for each AI-use category. Mean satisfaction is fairly stable across categories, ranging from 7.11 (monthly/infrequent users) to 7.30 (daily users), with non-users sitting at 7.13. The gradient from non-use to daily use is monotone but shallow. Figure 2 shows the distribution of professional work experience in the analysis sample. The distribution is right-skewed with a median of 11 years; roughly half of the sample has between 5 and 24 years of experience.

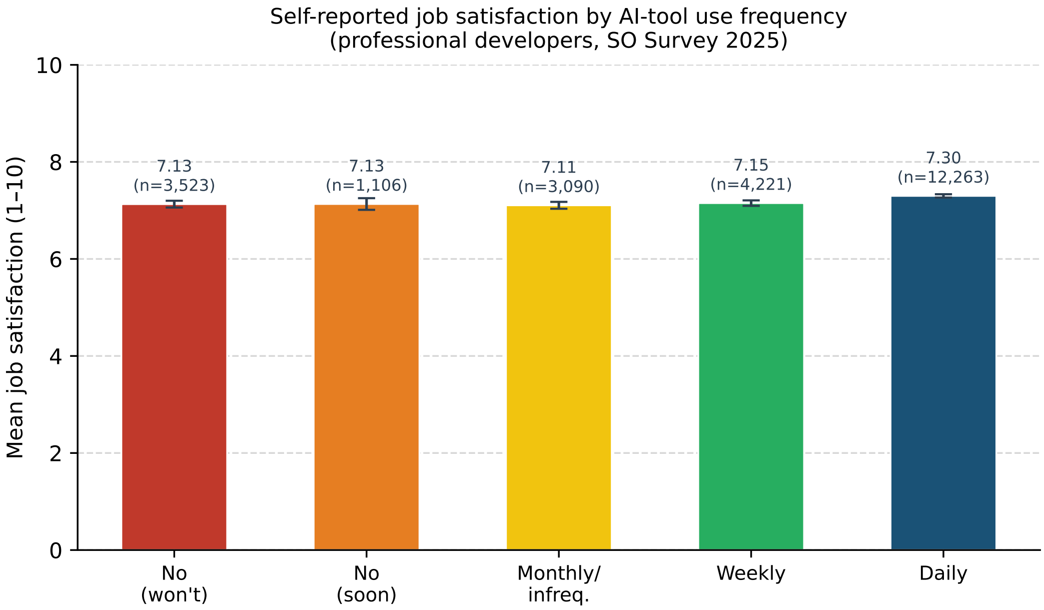

Figure 1: Mean self-reported job satisfaction (1–10 scale, with 95% confidence intervals) by AI-tool use frequency, restricted to professional developers with non-missing values on all analysis variables ($n = 24{,}203$). The bars are ordered from lowest to highest AI-use intensity. Numbers above each bar show the mean and respondent count for that category. Error bars are $\pm 1.96 \times SE$. Source: Stack Overflow Annual Developer Survey 2025.

4.2 Regression results

Table 1 reports coefficients on the key AI-use variable across the four models. In the bivariate regression (M1), AI-tool users report job satisfaction 0.107 points higher than non-users ($SE = 0.032$, $p < 0.001$). Adding work-experience controls (M2) increases the estimate slightly to 0.172 points, consistent with experience being negatively correlated with both AI-use adoption and job satisfaction (younger developers adopt more and report higher satisfaction on average). In the fully specified model with developer-role fixed effects (M3), the coefficient on binary AI use is 0.155 points ($SE = 0.033$, $p < 0.001$).

The ordinal specification (M4) yields a coefficient of 0.061 per unit step on the 0–4 AI-use scale ($SE = 0.009$, $p < 0.001$), consistent with the monotone pattern visible in Figure 1. Together, M3 and M4 suggest that *any* AI use is associated with roughly 0.15–0.17 points higher satisfaction, and that *more frequent* use is associated with slightly higher additional satisfaction.

The R^2 of the fully specified model is 0.020, indicating that AI-use frequency, work experience, and developer role jointly explain approximately 2% of the variance in self-reported job satisfaction. The association is real in the statistical sense but modest in practical terms: the AI-use premium

Figure 2: Distribution of years of professional work experience among the 24,203 respondents in the analysis sample. Values are capped at 40 years for display purposes; the dashed line marks the sample median of 11 years. Source: Stack Overflow Annual Developer Survey 2025.

of 0.155 points is an order of magnitude smaller than the inter-decile range of the satisfaction scale (approximately 5 points between the 10th and 90th percentile).

5 Limitations

The findings reported above carry six interconnected limitations, each of which prevents a causal reading of $\hat{\beta}_1$.

1. Non-random selection into the survey. The Stack Overflow Developer Survey reaches only developers who use or follow Stack Overflow, are aware of the survey, and choose to complete it. Developers with more positive job attitudes may be more or less likely to participate, biasing both the mean satisfaction level and its covariance with AI adoption in unknown directions. Any extrapolation to the full global developer workforce requires strong and untestable assumptions.

2. Self-reported AI-tool adoption. `AISelect` records what respondents say about their own AI-use frequency, not observed behaviour. Developers who are enthusiastic about AI tools may systematically over-report adoption frequency, introducing measurement error that is likely correlated with satisfaction and with technology-positive attitudes—a classic attenuation problem with a directional bias we cannot sign.

3. Self-reported job satisfaction. `JobSat` is a single-item ordinal measure of job satisfaction. Single-item satisfaction scales correlate with multi-item instruments but are noisier, and the exact interpretation of a one-point move on a 10-point scale varies across respondents and cultures. The international composition of the sample compounds this concern.

4. No instrument for AI-tool adoption. Establishing even a plausibly causal link between AI adoption and job satisfaction would require a source of quasi-random variation in adoption. No such instrument is available in these data. Developers who adopt AI tools likely differ from non-adopters in ability, curiosity, employer policy, team culture, and prior satisfaction—all

Table 1: OLS estimates: AI-tool use and self-reported job satisfaction. Dependent variable: job satisfaction (0–10). Robust standard errors in parentheses. Stars: ${}^{*}p < 0.05$, ${}^{**}p < 0.01$, ${}^{***}p < 0.001$.

	(M1) Bivariate	(M2) +Experience	(M3) +Role FE	(M4) Ordinal AI
Uses AI (binary)	0.107***	0.172***	0.155***	
	(0.032)	(0.032)	(0.033)	
AI use (0–4 scale)				0.061***
				(0.009)
Work experience		−0.026***	−0.027***	−0.027***
		(0.005)	(0.005)	(0.005)
Work experience^2		0.001***	0.001***	0.001***
		(0.000)	(0.000)	(0.000)
Developer-role FE	No	No	Yes	Yes
Observations	24,203	24,203	24,203	24,203
R^2	0.0004	0.0130	0.0202	0.0212

variables that are plausibly correlated with both adoption and the outcome. The role fixed effects in (M3) absorb some of these confounders at the occupation level but cannot handle within-role heterogeneity.

5. Omitted variables. Two categories of omitted variable bias are especially plausible. First, *employer-level confounders*: high-quality employers may simultaneously encourage AI adoption and create conditions for higher satisfaction (e.g., better pay, autonomy, and management quality). Second, *individual ability and disposition*: high-ability or curiosity-driven developers may be early AI adopters and independently more satisfied with their work. Both channels would produce an upward bias in $\hat{\beta}_1$.

6. Cross-sectional design. The survey provides a single cross-section taken at one point in time. We cannot distinguish between AI adoption raising satisfaction, satisfied developers being more likely to adopt AI tools, or a third variable driving both. Panel data tracking the same developers before and after AI-tool adoption would be necessary, though not sufficient, to approach causal identification.

6 Conclusion

We document a small, statistically significant positive cross-sectional association between AI-tool adoption and self-reported job satisfaction among professional software developers in the 2025 Stack Overflow Developer Survey. After controlling for work experience and developer role, AI-tool users report satisfaction scores approximately 0.155 points (out of 10) higher than non-users. The association is monotone in AI-use frequency, survives across all four model specifications, and is precisely estimated ($p < 0.001$), but it explains only 2% of the total variance in satisfaction and is small relative to the natural dispersion of the outcome. We emphasise that this correlation is descriptive: the six limitations discussed in Section 5 collectively preclude any causal interpretation.

This paper was produced as a teaching artefact for a university seminar on AI for research. Its purpose is not to advance the literature on developer wellbeing—though we hope the clean descriptive finding offers a useful data point—but to demonstrate the full research loop: obtaining and licensing public data, running a pre-specified analysis, writing results into a structured manuscript, and managing the entire pipeline under version control with AI-assisted tooling. The workflow, prompt templates, and reproducible analysis code are available in the project repository alongside this document. We encourage seminar participants to identify the limitations above not as failures but as a model for honest empirical writing: the most important scientific contribution is knowing what your design cannot tell you.

References

Stack Overflow. Annual developer survey 2025. `https://survey.stackoverflow.co/`, 2025. Licensed under ODbL 1.0 and DbCL 1.0. Replace year if a different snapshot is used.